Randomized Command-Level Reversal of Signed Hippocampo-Cortical Propagation: A Causal Test of Reported Temporal-Order Balance Against Restricted Neural Baselines

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AI-generated speculative research notice. This manuscript was generated by an artificial-intelligence system in a consciousness-research simulation. It reports no completed experiment, participant enrollment, unpublished dataset, pilot result, safety result, or empirical effect. The proposed mechanism, mathematics, stimulation program, causal estimands, and decision rules are hypotheses for prospective development. The available sources support selected component ideas but do not establish that bidirectional hippocampo-cortical propagation can be safely induced, matched, and measured in humans. Any implementation would require independent engineering validation, clinical approval, statistical design analysis, participant consultation, ethics review, and staged feasibility evidence.

Abstract

We propose a staged causal test of whether randomized command-level reversal of signed neural propagation changes **reported temporal-order balance** between two distributed visual representations. The primary endpoint is not phenomenal simultaneity. It is the physical onset difference at which a participant is equally likely to report "B later than A" or "A later than B" in a binary temporal-order judgment. We call this the temporal-order-judgment order-balance point, or **TOJ-OBP**. Explicit simultaneity reports, detection, confidence, psychometric resolution, and lapse behavior are modeled separately.

The motivating neural hypothesis is that a transient propagation event supplies a location-dependent reference time. If $\tau_\Gamma(x)$ is the arrival time of a realized signed activation-delay pattern Γ at location x, then a local content event encoded at physical time e_i has a candidate relational coordinate $q_i = e_i - \tau_\Gamma(x_i)$. For two items, the model predicts an order-balance displacement proportional to the signed arrival lag

$$L_\Gamma = \tau_\Gamma(x_B) - \tau_\Gamma(x_A),$$

provided that stimulation does not produce an equal displacement through sensory latency, response criterion, or another directed neural pathway. A causal implementation is specified using a wave-triggered synaptic trace sampled by content encoding; the trace supplies the memory operation missing from a purely descriptive change of coordinates. The model generates timing-window predictions that differ from some direct sensory-latency accounts, although the proposed experiment cannot guarantee complete mechanistic separation.

The primary causal estimand is the effect of randomized **stimulation-sign assignment** on TOJ-OBP in participants and montages found eligible using behavior-blind, independent calibration data. A pre-assignment neural-state gate is applied before randomization. Confirmatory analyses retain all safely randomized trials with behavioral responses and do not condition on realized post-stimulation lag, power, coherence, decoding, recruitment, or wave quality. Realized propagation is an aggregate manipulation check and a secondary latent mediator, not a trial-inclusion criterion. Because reversal can alter many directed neural consequences besides arrival lag, randomized assignment is not assumed to be a valid instrument for lag.

The program begins with phantom tests, hardware-in-the-loop validation, artifact characterization, and nonconfirmatory physiological pilots. Sparse intracranial recordings can establish ordered activation across a prespecified anatomical graph; the stronger term *wavefront* is reserved for patterns satisfying continuity, velocity, reference-robustness, and held-out propagation criteria. The proposal is motivated by computational evidence that excitability gradients can generate hippocampal waves and direction-specific hippocampal–cortical interactions, but that work does not demonstrate human stimulation feasibility, temporal-order effects, or consciousness mechanisms [1]. Late distributed activity associated with reportability is treated as a restricted comparison variable rather than as consciousness itself [7].

A confirmatory result would establish a causal effect of assigned stimulation sign on structured temporal report under the tested conditions. A quantitative relation between latent arrival lag and report would provide additional model-based evidence, subject to measurement and mediation assumptions. Any inference to phenomenal temporal order would remain conditional on an independently defended bridge from convergent reports and behavior to experience.

Keywords: traveling waves; signed activation delay; hippocampus; posterior cortex; temporal-order judgment; order-balance point; intracranial stimulation; causal estimand; measurement error; representational similarity analysis

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Revision note and response to independent review

The revised manuscript accepts the reviewers' central causal objection. The original primary analysis admitted confirmatory trials only when post-stimulation neural responses met wave, equivalence, decoding, and artifact criteria. That strategy would condition on consequences of randomized stimulation and could select different latent network states in the two assignment arms. Randomization would no longer identify the total effect of assignment in the selected data. The primary estimand has therefore been replaced by an assignment-based intention-to-treat estimand in a population whose eligibility is determined before the confirmatory task using independent calibration data. The online state gate is evaluated before assignment. Realized lag and all other neural responses are manipulation checks or secondary mechanistic variables, not primary inclusion variables.

The manuscript also accepts the reviewers' measurement objection. A 50% crossing in a forced binary order judgment is not an unqualified point of subjective simultaneity. It is now called the TOJ order-balance point. Explicit simultaneity judgments receive a separate observation model, and the direct conclusion is restricted to reported order. Detection, confidence, response mapping, sequential effects, and temporal resolution are not treated as interchangeable manifestations of a single observed phenomenal variable.

The original co-moving transformation was underimplemented. The revision retains it as a candidate relational coordinate but adds a causal readout model: a local wave marker initiates a decaying trace, item encoding samples that trace, and a downstream comparator uses the sampled trace to estimate an interval relative to local wave arrival. This supplies an explicit memory kernel and connects local arrival times to a decision variable. It also produces a factorial prediction across stimulation timing windows. The model remains hypothetical; specifying an implementation does not show that the brain uses it.

The neural-field derivation has been narrowed. Stronger sufficient conditions are stated for the high-gain front calculation, including kernel positivity, symmetry, decay, a threshold condition involving $Q(0)$, profile monotonicity, and a unique positive-speed root. Smooth-gain continuation and finite-time stability are presented only under explicit Fredholm and semigroup assumptions. Reversing a mathematical path coordinate is no longer described as evidence that an asymmetric biological network supports physical reverse propagation.

The original numerical lag range, lifetime, decoding threshold, standardized equivalence margin, sample target, unit-slope tolerance, cross-validated R^2 threshold, and one-quarter falsification rule lacked empirical calibration. They have been removed as confirmatory constants. Final values must be set using independent engineering data and simulation-based operating-characteristic analysis. Equivalence margins must be expressed in native or reliability-calibrated units tied to plausible behavioral consequences. The design now has three outcome classes—support, refutation under operational scope, and inconclusive—rather than forcing every estimate into confirmation or falsification.

The comparison targets have also been narrowed. The experiment does not test recurrent ignition, synchrony, or clock theories as complete families. It compares explicit restricted models: scalar late recruitment, unsigned coupling, centralized common-time, local sensory-latency, and signed-arrival models. A signed-lag advantage would show that selected scalar and undirected predictors are insufficient; it would not refute recurrent access, phase-sensitive synchrony, or content-specific clock theories generally.

Some reviewer requests cannot be resolved by textual revision alone. No stored source demonstrates safe human bidirectional hippocampo-cortical wave synthesis, artifact-free arrival estimation during stimulation, or a causal effect on temporal-order report. The manuscript therefore no longer presents a confirmatory human experiment as immediately ready. It specifies a staged research program whose confirmatory phase is contingent on prior engineering and physiological evidence. Likewise, the causal effect of realized arrival lag is not identified merely by randomizing command sign. The exclusion restriction required for a conventional instrumental-variable analysis is doubtful because stimulation reversal can alter local excitability, pulse-order effects, phase-amplitude relations, directed recruitment, sensation, and sensory latency. We therefore decline to present an instrumental-variable estimate as the primary mechanistic solution. A latent-lag analysis is retained as secondary and explicitly assumption-dependent.

Finally, we preserve a limited disagreement about conditional falsification. Inability to create the intervention does not logically refute the mathematical statement “if a suitable signed propagation coordinate is realized, then a specified readout may use it.” It does, however, count against the practical applicability and neuroscientific reach of the proposal. The revised program therefore has finite feasibility stopping rules and treats repeated failure to achieve a safe, reproducible bidirectional contrast as an adverse result for the research program, even though it is not a clean test of the conditional readout equation.

1. Research aim and claim hierarchy

1.1 Primary research question

The primary question is:

> **Among participants and stimulation montages found eligible on independent calibration data, what is the causal effect of randomized positive-sign versus negative-sign stimulation assignment on the temporal-order-judgment order-balance point?**

This formulation deliberately refers to **stimulation assignment**, not to a post hoc subset of trials with successful realized waves. It also refers to **reported order balance**, not directly to phenomenal simultaneity.

The proposed intervention attempts to reverse ordered neural recruitment across two independently mapped posterior-cortical item ensembles and a prespecified cortical–hippocampal graph. It is called command-level reversal because the input sequence, rather than the complete biological trajectory, is under direct experimental control. Whether the neural response reverses is an empirical first-stage question.

1.2 Four distinct levels

The manuscript distinguishes four levels throughout:

1. **Physical stimulus timing.**

The experiment controls the onset difference

$$\delta = t_B - t_A,$$

where $\delta > 0$ means that item B physically begins later than item A.

1. **Neural timing.**

The experiment estimates the times at which an independently specified neural marker appears at mapped locations. For a realized propagation pattern Γ,

$$L_\Gamma = \tau_\Gamma(x_B) - \tau_\Gamma(x_A).$$

1. **Reported temporal order.**

The primary behavioral response is binary: “B later than A” versus “A later than B.” Separate tasks measure explicit simultaneity, detection, identity, and confidence.

1. **Phenomenal temporal order.**

This is the experienced relation of earlier, later, or together. It is not directly manipulated or observed. Behavioral reports constrain hypotheses about it but are also influenced by encoding, memory, criterion, uncertainty, and response selection.

A causal effect at level 3 does not by itself establish a causal effect at level 4. The strongest defensible direct claim is about reported temporal organization.

1.3 Claim hierarchy

The proposal has three inferential layers.

Layer I: randomized behavioral effect. Randomized stimulation-sign assignment changes, or does not change, the TOJ-OBP in the independently eligible population. This is the primary causal claim.

Layer II: model-based neural mechanism. The assignment effect is quantitatively associated with a latent signed arrival lag, survives explicit comparison with sensory-latency and restricted neural baselines, and follows timing-window predictions of a trace-based readout. This layer requires measurement and mediation assumptions beyond randomization.

Layer III: conditional phenomenal interpretation. Convergence across forced order, explicit simultaneity, confidence, nonverbal behavior, and independently predicted neural consequences may support an interpretation concerning experienced temporal order. Such convergence would constrain but not identify phenomenal structure.

Only Layer I is the primary confirmatory estimand. Layers II and III are progressively more assumption-dependent.

1.4 Why the question is worth testing

A traveling or sequentially propagating neural pattern carries signed information not captured by total power or unsigned coupling. If that signed information affects order reports while selected scalar quantities are stable, it would identify propagation direction as a behaviorally relevant intervention variable. The result would be useful even if the ultimate mechanism were a sensory-latency change rather than the proposed relational coordinate.

The stronger hypothesis is that a downstream circuit uses the relation between content-encoding time and local propagation-arrival time. That hypothesis is risky because it predicts a quantitative displacement, not merely a nonspecific behavioral disturbance. It is also vulnerable to clear alternatives: direct latency shifts, response criteria, stimulation sensations, attention, pulse-order effects, and unmeasured directed network changes.

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2. Evidential foundation and limits

2.1 Computational motivation for hippocampal and cortical propagation

A biophysically grounded computational study reports that spatial gradients in input or excitability can generate hippocampal traveling waves above a model-specific threshold. The modeled waves have biologically plausible velocities, can reverse direction across frequency regimes, and produce direction-specific hippocampal–cortical interactions. In that model, anterior-to-posterior and posterior-to-anterior waves are not equivalent in their cortical consequences, and cortical waves can also induce structured hippocampal activity [1].

This source supports the following limited propositions:

- excitability gradients can generate propagation in a specified model;
- propagation direction can matter to coupled hippocampal–cortical dynamics;

- cortical and hippocampal waves can interact bidirectionally in that model;
- opposite directions may have asymmetric downstream effects.

It does not show:

- that the proposed waves occur during visual temporal-order judgments;
- that human intracranial stimulation can safely reverse them;
- that forward and reverse responses can be matched on other directed consequences;
- that sparse contacts can verify a continuous cortical–hippocampal front;
- that propagation affects temporal report or consciousness.

The modeled directional asymmetry is not merely supportive. It is also a reason to doubt that matched reverse propagation will be feasible. The engineering program must test that doubt rather than assume it away.

2.2 Reportability and late distributed activity

In a backward-masking paradigm, substantial early occipito-temporal processing occurred below approximately 250 ms, while later distributed fronto-parieto-temporal activity after approximately 270 ms correlated with subjective and objective reportability [7]. This finding motivates a distinction between early content processing and later report-related recruitment.

It does not establish that late distributed activity is consciousness itself, that contact count is a sufficient statistic for recurrent ignition, or that a fixed time window transfers to the present task. Accordingly, the revised protocol does not use a generic 250–500 ms contact count as a validated assay of conscious access. A task-specific late-recruitment measure must be defined from independent localizer data, and it is treated as one restricted comparison variable.

2.3 Temporal-order and simultaneity measurements

Prior work has addressed determinants of simultaneity and order perception [12]. A conference record reports that visual N1 latency predicted individual location of a point of subjective simultaneity in audiovisual temporal-order judgments, although the stored record is too sparse to establish effect size, design quality, or transfer to the present intracranial visual task [11]. Other work explicitly distinguishes sequential dependence in subjective temporal-order reports from subjective simultaneity [14]. A tactile rubber-hand-illusion study also reports distinct modulation of psychometric center and temporal resolution, but that result is modality- and paradigm-specific [13].

These records justify separating:

- forced binary order balance;
- explicit simultaneity reports;
- temporal resolution;
- sequential dependence;
- detection and lapse behavior.

They do not justify treating any one measure as a transparent readout of phenomenal simultaneity.

2.4 Representational analysis and intracranial coverage

Representational similarity analysis compares relational geometry among multichannel neural patterns, models, and behavior using representational dissimilarity matrices [6]. It is relevant because electrode identities do not correspond straightforwardly across participants and because the hypothesis concerns distributed relations.

Variable electrode placement is a central problem in multi-patient intracranial research. Available methodological families include generalized linear models, multivariate pattern analysis, representational similarity analysis, hierarchical factorization, Gaussian processes, geometric alignment, and inter-subject comparisons [16]. These tools do not themselves solve causal selection, arrival-time measurement error, or temporal-warp overfitting. The revised design therefore privileges within-participant randomization, independently fixed paths, and cross-fitted representational analyses.

2.5 Closed-loop systems

Chronically implanted cortical sensing has been used to enable or disable deep-brain stimulation according to decoded movement state in essential tremor [20]. This supports only broad feasibility of combining implanted sensing with contingent stimulation. It does not support multichannel hippocampo-cortical propagation synthesis, artifact-free arrival estimation during stimulation, or manipulation of temporal-order reports.

2.6 Novelty and priority

The proposed contribution is the conjunction of:

1. independently calibrated command-sign reversal;
2. randomized assignment in a pre-eligible population;
3. measured signed activation-delay structure;
4. separate psychometric models for order and simultaneity;
5. an explicit causal trace readout;
6. cross-fitted comparison of physical-time and relational neural geometry.

The coordinate transformation $t - \tau(x)$, Fourier conjugation under ideal time reversal, traveling-front reduction, and representational similarity analysis are not claimed as individually novel. The stored source set does not contain a sufficient nearest-neighbor review of causal traveling-wave manipulation, phase-gradient stimulation, latency compensation, or propagation effects on perception. Priority therefore remains unestablished. A systematic review of that literature is a prerequisite for any strong novelty claim.

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3. Dynamical scaffold for signed propagation

3.1 Role of the neural-field model

The field model is a scaffold for intervention design. It is not identified with consciousness, and it is not treated as a validated model of each participant's anatomy. Its purposes are to:

- define a signed propagation coordinate;
- clarify conditions under which a front-like solution can exist;

- distinguish existence from finite-time stability;
- show why opposite command patterns need not produce equivalent biological responses;
- motivate measurable arrival-time and path-level criteria.

3.2 Coupled cortical–hippocampal field

Let M_C and M_H denote effective cortical and hippocampal domains. For receiving subsystem $a \in \{C, H\}$, define activity $u_a(x,t)$ at location $x \in M_a$ and physical time t. Consider

$$\tau_a(x)\,\partial_t u_a(x,t) = -u_a(x,t) + \sum_{b\in\{C,H\}} \int_{M_b} W_{ab}(x,y) f_b(u_b(y,t))\ d\mu_b(y) + g_a(x) + I_a(x,t) \quad (1)$$

Here:

- $\tau_a(x) > 0$ is a local effective time constant;
- $W_{ab}(x,y)$ is an effective directed coupling kernel;
- f_b is a bounded population-gain function;
- μ_b is a spatial or graph measure;
- $g_a(x)$ is baseline drive or excitability;
- $I_a(x,t)$ is experimental input.

The domains may be graph-embedded rather than continuously sampled surfaces. In sparse human recordings, Equation 1 is at most a coarse latent-field model.

3.3 Assumptions

The formal development uses the following assumptions.

D1. Bounded dynamics. The gain functions are bounded, and the coupling operators are bounded on the selected function space.

D2. Effective path. A prespecified path or anatomical graph crosses the sampled posterior-cortical sites and hippocampal contacts. The path is fixed without confirmatory behavioral data.

D3. Excitable contrast. The reduced dynamics support low- and high-activity regimes, or a pulse with a plateau long enough to estimate ordered arrival.

D4. Direction-selecting structure. Anatomical coupling, excitability gradients, or experimental input can bias propagation sign.

D5. Observable marker. An independently specified and artifact-validated marker provides probabilistic information about local arrival time.

D6. Finite-time reproducibility. The pattern is sufficiently stable over a task-relevant interval to yield held-out arrival predictions. No claim of eternal or asymptotic stability is required.

D7. Sparse-data restraint. A continuous wavefront is inferred only if continuity and propagation criteria are met. Otherwise the observation is called a signed activation-delay map.

Assumptions D5–D7 are empirical requirements, not consequences of Equation 1.

4. A sufficient high-gain front calculation

4.1 One-dimensional reduction

Let $\rho \in \mathbb{R}$ denote distance along a candidate path. A scalar reduction of Equation 1 is

$$\tau_0 \partial_t u(\rho, t) = -u(\rho, t) + \int_{\mathbb{R}} w(\rho - \rho') f(u(\rho', t))\ d\rho' + g_0, \tag{2}$$

where $\tau_0 > 0$, w is a translation-invariant kernel, and g_0 is constant drive.

Seek a profile

$$u(\rho, t) = U(\xi), \qquad \xi = \rho - ct, \tag{3}$$

where c is signed speed. Substitution gives

$$\tau_0 c U'(\xi) - U(\xi) + \int_{\mathbb{R}} w(\xi - \eta) f(U(\eta))\ d\eta + g_0 = 0. \tag{4}$$

4.2 High-gain assumptions

For an explicit sufficient calculation, assume:

1. $f(u) = H(u - \theta)$, where H is the Heaviside function;
2. $w \in L^1(\mathbb{R})$ is continuous almost everywhere;
3. $w(r) \geq 0$;
4. w is even, $w(r) = w(-r)$;
5. w is not identically zero and decays sufficiently for the following integrals to exist.

Define

$$\kappa = \int_{\mathbb{R}} w(r)\, dr \tag{5}$$

and

$$Q(\xi) = \int_{\xi}^{\infty} w(r)\, dr. \tag{6}$$

Evenness gives

$$Q(0) = \frac{\kappa}{2}. \tag{7}$$

Assume a descending front with $U(\xi) > \theta$ for $\xi < 0$ and $U(\xi) < \theta$ for $\xi > 0$. Then the active half-line is $\eta < 0$, and Equation 4 becomes

$$\tau_0 c U'(\xi) - U(\xi) + Q(\xi) + g_0 = 0. \tag{8}$$

For $c > 0$, the bounded solution is

$$U_c(\xi) = \frac{1}{\tau_0 c} \int_{\xi}^{\infty} \exp\left[-\frac{r-\xi}{\tau_0 c}\right] [Q(r) + g_0] \, dr. \tag{9}$$

The interface condition $U_c(0) = \theta$ requires

$$\theta = F(c) = g_0 + \frac{1}{\tau_0 c} \int_0^{\infty} e^{-r/(\tau_0 c)} Q(r) \, dr. \tag{10}$$

4.3 Existence and uniqueness of a positive-speed root

Under the assumptions above,

$$\lim_{c \downarrow 0} F(c) = g_0 + Q(0) = g_0 + \frac{\kappa}{2}, \tag{11}$$

whereas

$$\lim_{c \to \infty} F(c) = g_0. \tag{12}$$

Because $Q(r)$ is nonincreasing and nonconstant, increasing the exponential averaging scale $\tau_0 c$ shifts weight toward larger r, where $Q(r)$ is smaller. Thus $F(c)$ is strictly decreasing under the stated regularity conditions. A unique positive root therefore exists if

$$g_0 < \theta < g_0 + Q(0) = g_0 + \frac{\kappa}{2}. \tag{13}$$

This is stronger than the insufficient condition $g_0 < \theta < g_0 + \kappa$.

Moreover, $Q(\xi) + g_0$ is nonincreasing. Equation 9 is a forward exponential average of that nonincreasing function, so $U_c(\xi)$ is nonincreasing. Its asymptotic values are

$$\lim_{\xi \to -\infty} U_c(\xi) = g_0 + \kappa, \qquad \lim_{\xi \to \infty} U_c(\xi) = g_0. \tag{14}$$

With strict monotonicity and $U_c(0) = \theta$, the assumed single crossing is self-consistent.

These statements are sufficient only for the idealized nonnegative symmetric kernel. Sign-changing, asymmetric, delayed, or finite-domain kernels require separate analysis. In particular, anatomical asymmetry may change existence, speed, recruitment, and stability in ways not captured by reflection of ρ.

4.4 Reversal is not established by coordinate reflection

Replacing ρ with $-\rho$ changes the coordinate description. It does not show that the same biological network supports a physically reversed front under an equivalent input. A reverse biological response may require a different drive, may recruit different pathways, or may not exist.

The motivating computational work reports direction-specific hippocampal–cortical consequences rather than symmetry [1]. Therefore, bidirectional feasibility and downstream similarity are empirical questions. The mathematical front calculation supplies no license to assume them.

4.5 Smooth-gain continuation

Let f_ε be a family of smooth sigmoids converging to the Heaviside function as $\varepsilon \downarrow 0$. Persistence of the front is not automatic. A valid continuation argument would require:

- a specified weighted function space;
- a traveling-profile operator differentiable in U, c, and ε;
- a linearized operator that is Fredholm of index zero;
- a one-dimensional kernel generated by translation;
- an adjoint null vector with nonzero pairing to the speed derivative;
- a phase condition removing the translation mode;
- invertibility of the augmented operator.

Under those assumptions, an implicit-function argument may continue the pair (U, c) for sufficiently small ε . This manuscript does not prove that participant-specific neural fields satisfy those assumptions. The smooth-gain front remains a model template.

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5. Finite-time stability and coupled phase relations

5.1 Semigroup assumption

Let $\mathcal{L}$ be the linearized generator around a fitted front in a Banach space X. Let P project away from the translational mode. A spectral statement alone is insufficient, especially for non-normal or infinite-dimensional operators. The required assumption is the semigroup bound

$$\left\| e^{\mathcal{L}t} P \right\|_{X \to X} \le C_s e^{-\alpha t}, \qquad t \ge 0, \tag{15}$$

where $C_s \ge 1$ permits transient amplification and $\alpha > 0$ is a decay rate.

If the projected perturbation $v(t)$ obeys

$$\partial_t v = \mathcal{L} v + r(t)$$

with $\|r(t)\|_X \le \epsilon_r$, Duhamel's formula gives

$$\|v(t)\|_X \le C_s e^{-\alpha t} \|v(0)\|_X + \frac{C_s \epsilon_r}{\alpha} \left(1 - e^{-\alpha t}\right). \tag{16}$$

Equation 16 is conditional on Equation 15. It is not inferred from a spectral plot alone. In the empirical protocol, finite-time stability is assessed directly through held-out propagation prediction and residual bounds rather than inferred from an unverified operator spectrum.

5.2 Operational meaning of metastability

The term *metastable* is used only if a pattern:

- persists beyond the minimum duration required by the task;
- retains direction and approximate shape;
- permits held-out prediction of sampled-node arrival times;
- eventually terminates, leaves the sampled graph, or transitions after stimulation.

The duration is not fixed in advance at an unsupported universal value. It is determined by engineering and physiological pilots, then frozen before confirmatory data collection. It is not interpreted as the duration of a "specious present."

5.3 Coupled cortical and hippocampal phase equations

Let $a_C(t)$ and $a_H(t)$ be reduced cortical and hippocampal pattern positions. A generic weak-coupling reduction is

$$\dot{a}_C = c_C + \lambda \mathcal{H}_{CH}(a_H - a_C), \tag{17}$$

$$\dot{a}_H = c_H + \lambda \mathcal{H}_{HC}(a_C - a_H). \tag{18}$$

For relative displacement $d = a_H - a_C$,

$$\dot{d} = (c_H - c_C) + \lambda \left[\mathcal{H}_{HC}(-d) - \mathcal{H}_{CH}(d)\right]. \tag{19}$$

A locked offset d^* satisfies

$$(c_H - c_C) + \lambda \left[\mathcal{H}_{HC}(-d^*) - \mathcal{H}_{CH}(d^*)\right] = 0, \tag{20}$$

and is locally stable if the derivative of the right-hand side with respect to d is negative at d^*.

These equations are generic. They do not generate a quantitative cortical–hippocampal prediction until the interaction functions are estimated or constrained. The confirmatory behavioral theory does not rely on a fitted value of d^*. The equations serve only to emphasize that cortical and hippocampal patterns may lock in one direction but not the other.

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6. Command-level reversal and realized neural timing

6.1 Ideal continuous-signal reversal

Let $s_k(t)$ be a finite real-valued command signal at contact k, zero outside $0 \le t \le T$. Its ideal temporal reverse is

$$s_k^R(t) = s_k(T - t). \tag{21}$$

Under the Fourier convention

$$S_k(f) = \int_{-\infty}^{\infty} s_k(t) e^{-i2\pi f t}\, dt,$$

the reversed signal satisfies

$$S_k^R(f) = e^{-i2\pi f T} S_k^*(f). \tag{22}$$

Consequently,

$$|S_k^R(f)|^2 = |S_k(f)|^2. \tag{23}$$

If

$$G_{kl}(f) = S_k(f) S_l^*(f)$$

is a pairwise cross-spectrum, then

$$G_{kl}^R(f) = G_{kl}^*(f), \tag{24}$$

so

$$|G_{kl}^R(f)| = |G_{kl}(f)|. \tag{25}$$

Thus ideal common reversal preserves contact-wise power spectra and pairwise cross-spectral magnitudes while reversing cross-spectral phase. It does not preserve higher-order temporal statistics, waveform asymmetry, cross-frequency coupling, adaptation, biological response, or every perceptible feature.

6.2 Hardware-constrained reversal

Literal sample-by-sample reversal can reverse the order of phases within a biphasic pulse and may violate required pulse shape or electrode-polarization constraints. The implementable intervention must therefore be distinguished from Equation 21.

The proposed hardware operation is **event-sequence reversal**:

1. define a clinically approved charge-balanced pulse primitive;
2. preserve the within-pulse phase order and width;
3. reverse the order and inter-contact timing of pulse events;
4. preserve contact-wise total charge and pulse count;
5. verify actual output on a hardware load;
6. quantify rather than assume deviations from ideal spectral invariance.

The implemented pair is therefore a constraint-preserving approximation to ideal reversal. Any spectral or temporal matching claim applies only to measured command outputs and, separately, to measured neural responses.

6.3 Phase-rotated controls

For a multichannel Fourier vector $\mathbf{S}(f)$, let the target cross-spectral matrix be

$$\mathbf{G}(f) = \mathbb{E}\left[\mathbf{S}(f)\mathbf{S}(f)^{\dagger}\right]. \tag{26}$$

Let $\mathbf{D}(f)$ be diagonal with entries $e^{i\phi_k(f)}$. A phase-rotated target is

$$\mathbf{G}'(f) = \mathbf{D}(f)\mathbf{G}(f)\mathbf{D}(f)^{\dagger}. \tag{27}$$

This preserves diagonal power and pairwise cross-spectral magnitudes algebraically. To generate real commands, the rotations must obey conjugate symmetry,

$$\phi_k(-f) = -\phi_k(f). \tag{28}$$

The constructed waveform must also satisfy charge balance, amplitude, pulse-width, duty-cycle, timing-resolution, and hardware constraints. Preservation of Equation 27 does not imply equivalent neural responses.

6.4 Arrival time as a latent quantity

For trial j and sampled node k, let τ_{jk} denote latent arrival time. Neural measurements may include phase crossings, envelope changes, high-frequency activity, or latent-state transitions. For measurement family m,

$$\widehat{\tau}_{jkm} = a_{km} + b_{km}\tau_{jk} + \epsilon_{jkm}, \tag{29}$$

where a_{km} is a marker-specific offset, b_{km} is a scale parameter, and ϵ_{jkm} is measurement error. Their calibration distributions must be estimated without confirmatory behavior.

For posterior-cortical item centroids x_A and x_B,

$$L_j = \tau_j(x_B) - \tau_j(x_A) \tag{30}$$

is the latent signed lag. The item centroids are cortical representations established by an independent localizer. The protocol does not assume hippocampal retinotopy. Hippocampal contacts contribute to the larger propagation graph, not to the definition of retinotopic item location.

6.5 Crossing rules and multiple events

The primary arrival estimator must specify:

- marker identity;
- crossing or state-transition direction;
- minimum signal-to-noise ratio;
- treatment of multiple crossings;
- treatment of missing crossings;
- contact-specific uncertainty;
- path uncertainty;
- pulse-contaminated intervals;
- behavior-blind quality control.

A reverse condition may invert the temporal derivative of a marker. The estimator therefore uses a **direction-conjugate state-transition rule** fixed during calibration: the same latent front state is identified under opposite graph orientation, rather than applying one arbitrary voltage-crossing direction to both conditions. If the mapping between marker direction and latent arrival cannot be validated in synthetic and phantom data, the arrival analysis is not interpretable.

6.6 Wavefront versus signed activation-delay map

A pattern is called a **signed activation-delay map** if sampled nodes show reproducible ordered timing along the prespecified graph.

It is called a **wavefront** only if additional evidence supports:

- local continuity across adjacent sampled or interpolated nodes;
- a coherent velocity range;
- held-out prediction of omitted-node arrivals;
- persistence across reference schemes;
- physiological features outside pulse-locked artifact;
- failure of common-input and disconnected-channel controls;
- residual structure compatible with propagation rather than arbitrary sequential recruitment.

Sparse electrodes cannot prove a continuous physical transition between cortex and hippocampus. Even a qualified wavefront inference remains model-based.

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7. From a coordinate transformation to a causal readout

7.1 The descriptive coordinate

Suppose a local content event i is neurally encoded at physical time

$$e_i = t_i + \ell_i, \tag{31}$$

where t_i is external onset and ℓ_i is sensory and local encoding latency. A propagation-relative coordinate is

$$q_i = e_i - \tau_\Gamma(x_i). \tag{32}$$

For two events,

$$q_B - q_A = (t_B - t_A) + (\ell_B - \ell_A) - [\tau_\Gamma(x_B) - \tau_\Gamma(x_A)]\,. \tag{33}$$

Using $\delta = t_B - t_A$ and $L_\Gamma = \tau_\Gamma(x_B) - \tau_\Gamma(x_A)$,

$$q_B - q_A = \delta + \ell_B - \ell_A - L_\Gamma. \tag{34}$$

Equation 32 alone is a reparameterization. It becomes a neural hypothesis only when a causal system generates and reads a variable related to q_i.

7.2 Common representational space

Let local content activity $z_i(t)$ lie in feature space V_i. Let

$$E_i : V_i \to V$$

be a specified embedding into a common downstream feature space V. Given fixed arrival times, a descriptive alignment operator is

$$\mathcal{B}_\Gamma[\mathbf{z}](s) = \sum_{i=1}^{n} w_i E_i z_i(s + \tau_\Gamma(x_i))\,, \qquad \sum_i w_i = 1. \tag{35}$$

The embeddings resolve the original type problem: vectors from different local spaces are not added until they are mapped into a common space. Equation 35 is useful for representational analysis, but it still does not show that a biological circuit performs temporal realignment.

7.3 A wave-triggered trace implementation

Consider a local propagation marker $g_i(t)$ concentrated around arrival $\tau_i = \tau_\Gamma(x_i)$. Let a causal trace $r_i(t)$ obey

$$\dot{r}_i(t) = -\frac{1}{\kappa_i} r_i(t) + A_i g_i(t), \qquad r_i(-\infty) = 0, \tag{36}$$

where $\kappa_i > 0$ is a decay constant and A_i is normalized marker amplitude.

For an ideal impulse marker $g_i(t) = \delta_D(t - \tau_i)$, where δ_D is the Dirac impulse,

$$r_i(t) = A_i e^{-(t-\tau_i)/\kappa_i} H(t - \tau_i). \tag{37}$$

Suppose content encoding at time e_i gates or samples this trace:

$$c_i = r_i(e_i). \tag{38}$$

When $e_i \geq \tau_i$,

$$c_i = A_i e^{-(e_i-\tau_i)/\kappa_i}. \tag{39}$$

If A_i and κ_i are calibrated or locally normalized, a downstream monotone decoder can estimate

$$\hat{q}_i = -\kappa_i \log\left(\frac{c_i}{A_i}\right) \approx e_i - \tau_i. \tag{40}$$

The nervous system need not implement an explicit logarithm. Over a restricted interval, recurrent or synaptic weights can approximate a monotone mapping from trace strength to elapsed time. The essential causal elements are:

- a wave-triggered local state;
- persistence of that state;

- sampling by content encoding;
- a downstream comparison across locations.

A comparator then forms

$$d_{BA} = \hat{q}_B - \hat{q}_A. \tag{41}$$

A simple report model is

$$\Pr(Y = 1 \mid d_{BA}) = \operatorname{logit}^{-1}\left[\beta(d_{BA} - \rho)\right], \tag{42}$$

where $Y = 1$ means “B reported later than A,” $\beta > 0$ is response sensitivity, and ρ is an order criterion.

Increasing δ increases d_{BA} and therefore increases the probability of reporting B later. The 50% balance point is

$$\theta_{\mathrm{TOJ}} = L_\Gamma + \ell_A - \ell_B + \rho. \tag{43}$$

Under stable sensory-latency difference and stable criterion,

$$\Delta\theta_{\mathrm{TOJ}} \approx \Delta L_\Gamma. \tag{44}$$

Equation 44 is the structural unit-slope prediction. It is not assumed to hold for noisy point estimates of lag, and it is not the primary causal estimand.

7.4 Alternative causal implementations

The trace in Equation 36 is one implementation, not the only one. Other possibilities include:

- a content trace sampled by later propagation;
- delay-compensated downstream projections;
- recurrent states invariant along a propagation characteristic;
- paired synaptic eligibility traces encoding content–wave interval;
- a buffer that retains local item states until a common readout.

These implementations need not make identical predictions. A post-encoding wave could affect report in a content-trace implementation but not in the strict wave-before-content implementation of Equations 36–40.

7.5 Factorial timing predictions

The stimulation program therefore varies propagation timing relative to item encoding:

1. **Pre-encoding condition.**

The wave marker arrives before local item encoding. The wave-triggered trace model predicts a measurable effect if the trace remains within its calibrated dynamic range.

1. **Encoding-overlap condition.**

Propagation overlaps sensory encoding. Both the trace model and a direct sensory-latency account may predict effects.

1. **Post-encoding, pre-response condition.**

Item-specific local activity is already established before propagation. The strict wave-triggered trace model predicts little or no effect because content did not sample a prior wave trace. A content-buffer implementation may predict an effect; a direct early sensory-latency account predicts none.

1. **Post-decision control.**

Stimulation occurs after the response-relevant comparison should be complete. A behavioral shift here would indicate carryover, criterion, motor, or nonspecific effects.

The interaction between assigned sign and timing window is secondary but mechanistically important. It prevents the coordinate account from being insulated against every timing result.

7.6 Remaining ambiguity with sensory latency

A wave may alter ℓ_A or ℓ_B. That pathway is biologically real and could generate the same TOJ displacement as Equation 44. Statistical adjustment for measured latency does not automatically isolate a direct coordinate effect because latency is post-treatment and measured with error.

The revised manuscript therefore treats latency change as a competing mediator, not as a nuisance that can simply be “controlled away.” The strongest coordinate-compatible pattern would combine:

- an assignment effect on TOJ-OBP;
- a signed latent-lag relation;
- no commensurate change in local content-onset latency;
- timing-window specificity predicted by a trace implementation;
- improved relational geometry after cross-fitted arrival correction.

If local sensory-latency displacement quantitatively accounts for the report shift, the coordinate interpretation is not supported, even if the randomized intervention changes reported order.

8. Primary causal estimand

8.1 Independently eligible population

Eligibility is determined before the confirmatory behavioral phase using separate, behavior-blind calibration data. A participant–montage pair is eligible if:

- the clinical team permits stimulation;
- two posterior-cortical item ensembles are independently localized;
- both items are decodable under no-stimulation conditions;
- both command signs can be delivered safely;
- aggregate calibration responses show reproducible opposite signed activation-delay patterns;
- artifact characterization permits interpretable neural measurement;
- no direction is systematically associated with unsafe afterdischarge;
- the confirmatory path, marker, stimulation library, and analysis code are frozen.

Eligibility may depend on calibration responses to both command signs. This restricts external validity to a wave-responsive population, but it does not condition the subsequent randomized confirmatory comparison on post-assignment outcomes.

8.2 Assignment and potential outcomes

Let $D \in \{-1, +1\}$ denote randomized command-sign assignment. Let $Y(d, \delta)$ be the potential binary order report under assignment d and physical onset difference δ, with $Y = 1$ meaning “B reported later than A.”

Define the assignment-specific population response curve

$$m_d(\delta) = \mathbb{E}[Y(d, \delta)] \,, \tag{45}$$

where the expectation averages over the independently eligible population, prespecified waveform-pair distribution, sessions, item identities, and pre-assignment states admitted by the online gate.

Assuming $m_d(\delta)$ is monotone increasing with a unique 50% crossing, define

$$\theta_d = \inf\left\{\delta : m_d(\delta) \geq \frac{1}{2}\right\}. \tag{46}$$

The primary causal estimand is

$$\Delta_{\text{ITT}} = \theta_{+1} - \theta_{-1}. \tag{47}$$

This is the effect of randomized command-sign assignment on TOJ order balance. It is not the causal effect of realized lag.

8.3 Identification assumptions

Randomization identifies Equation 47 under:

- **consistency:** assignment corresponds to the frozen waveform library;
- **positivity:** both signs have positive assignment probability at relevant states and stimulus timings;
- **pre-assignment gating:** the neural-state gate is evaluated before D is drawn;
- **no unmodeled interference:** one trial’s assignment does not alter another trial’s outcome after the specified washout and carryover terms;
- **outcome observation:** missing reports are handled under prespecified assumptions and sensitivity bounds;
- **correct standardization:** adaptive stimulus allocation and waveform-pair probabilities are included in estimation.

Carryover is plausible in neural stimulation. Trial spacing, prior assignment, prior response, and session time must therefore be modeled. Persistent aftereffects that invalidate the no-interference window require longer washout or a redesign.

8.4 Intention-to-treat inclusion

After the pre-assignment state gate passes, sign is randomized. The primary analysis includes every safely randomized trial with an elicited order response, regardless of:

- realized lag;
- wave classification;
- power;
- coherence;
- item decodability;
- recruitment;
- stimulation sensation;
- marker quality;
- whether neural equivalence was achieved on that trial.

Behavior remains part of the primary analysis even if neural recording is unusable. Trials are not deleted because the manipulation "failed."

Post-assignment absence of a behavioral response is treated as missing, not silently excluded. The protocol reports missingness by arm, uses prespecified models based on pre-assignment and device variables, and presents tipping-point or worst-case bounds. Safety-triggered termination is also reported as an intervention outcome.

8.5 Why post-treatment equivalence is not an inclusion criterion

Power, coherence, sensory latency, decodability, late recruitment, and artifact morphology can all be affected by assignment. Conditioning on them would change the estimand and could create selection bias. Their roles are therefore:

- interpretation of the biological contrast;
- restricted model comparison;
- safety monitoring;
- secondary heterogeneity analysis;
- evidence for or against the proposed mechanism.

They do not determine confirmatory trial membership.

8.6 Multiple independently randomized patterns

A single forward–reverse pair may confound sign with pulse order, local onset phase, or contact-specific effects. The stimulation library therefore contains multiple independently calibrated waveform pairs. Within each pair, sign is randomized. Across pairs, the design varies:

- absolute commanded delay;
- train-to-stimulus timing;
- local onset phase;
- contact-order realization;
- pulse-spacing pattern;

- graph segment emphasized.

The distribution of pairs is fixed before confirmatory enrollment. The primary estimand averages over that distribution. Pattern-specific estimates test whether the assignment effect generalizes rather than depending on one waveform.

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9. Why randomized sign does not identify a causal lag effect

9.1 Post-treatment mediator structure

Let:

- D be randomized command sign;
- L be latent realized arrival lag;
- M be other directed neural consequences;
- S be stimulation sensation;
- E be sensory-encoding latency;
- Y be order report.

Plausible pathways include

$$D \to L \to Y,$$

but also

$$D \to M \to Y, \qquad D \to S \to Y, \qquad D \to E \to Y.$$

Pretrial state can affect L, M, E, and Y. Randomization identifies the total effect of D, not the isolated effect of L.

9.2 Instrumental-variable assumptions

Using D as an instrument for L would require:

1. **relevance:** assignment changes latent lag;
2. **exclusion:** assignment affects report only through lag;
3. **independence:** assignment is randomized;
4. **monotonicity or another identified structural restriction;**
5. **no interference;**
6. **a valid measurement model for lag.**

Independence is plausible by design, and relevance can be tested. Exclusion is doubtful. Reversal can change local phase, effective connectivity, phase-amplitude coupling, adaptation, stimulation sensation, artifact morphology, and sensory latency. Matching a finite list of scalar summaries does not establish exclusion.

The confirmatory manuscript therefore does not label an instrumental-variable estimate as the causal effect of lag. If such an estimate is reported, it must be explicitly conditional on the exclusion restriction and accompanied by sensitivity analyses allowing bounded direct effects of assignment.

9.3 Secondary latent-lag model

A secondary joint model estimates latent lag from multiple markers and relates it to report:

$$\widehat{\mathbf{L}}_j \sim p(\widehat{\mathbf{L}}_j \mid L_j^*, \boldsymbol{\psi}), \tag{48}$$

$$\Pr(Y_j = 1) = \operatorname{logit}^{-1}\left[\beta_p\left(\delta_j - \alpha_p - b_L L_j^*\right) + \mathbf{x}_{j,\mathrm{pre}}^{\top}\boldsymbol{\gamma}\right]. \tag{49}$$

Here L_j^* is latent lag, $\boldsymbol{\psi}$ contains calibration-derived measurement parameters, and $\mathbf{x}_{j,\mathrm{pre}}$ contains only pre-assignment precision covariates in the primary form.

The structural trace model predicts $b_L = 1$ when:

- lag and report are expressed in the same time units;
- sensory-latency difference is stable;
- response criterion is stable;
- the readout is locally linear in corrected time;
- the correct arrival variable is measured;
- no additional gain rescales the relation.

The analysis propagates uncertainty in both lag and psychometric location. A regression on plug-in lag estimates is not used as the principal mechanistic test.

Even with this measurement model, b_L is an associational or structural parameter unless stronger mediation assumptions are defended.

9.4 Position on direct-effect adjustment

The primary total-effect model does not adjust for post-treatment latency, power, recruitment, or sensation. Secondary models may compare coordinate and latency pathways, but their estimands must be named.

For example, conditioning on measured sensory latency estimates a form of direct association, not automatically a causal direct effect. A controlled direct effect would require an intervention fixing latency to a specified value, which the present experiment does not provide. An interventional indirect effect would require assumptions about mediator distributions and mediator–outcome confounding. The manuscript preserves this limitation rather than treating adjustment as a solution.

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10. Staged feasibility and stopping program

10.1 Stage 0: literature, engineering, and regulatory preparation

Before participant stimulation, the program requires:

- a systematic nearest-neighbor literature review;

- specification of the pulse generator's exact output constraints;
- electromagnetic and thermal modeling where clinically required;
- a failure-mode and effects analysis;
- preregistered definitions of artifact, technical failure, and safety events;
- clinical and participant input into acceptable burden;
- a simulation-based design analysis.

No behavioral confirmation threshold is set before these steps are complete.

10.2 Stage 1: phantom and hardware-in-the-loop tests

A physical phantom and hardware load are used to measure:

- delivered current and voltage;
- contact-specific timing jitter;
- amplifier recovery;
- cross-channel leakage;
- reference dependence;
- pulse blanking requirements;
- differences between ideal and constraint-preserving reversal;
- false propagation produced by hardware alone.

Synthetic neural signals with known delays are added to recorded artifact. The complete arrival-estimation pipeline must recover those delays without using behavioral labels. Disconnected channels and non-neural sensors serve as negative controls.

A waveform pair fails if a direction decoder can classify sign from artifact-only intervals or disconnected hardware channels after the planned preprocessing.

10.3 Stage 2: nonconfirmatory physiological pilot

Where clinically and ethically permissible, a nonbehavioral pilot asks whether low-risk multichannel patterns produce reproducible ordered physiological responses. This stage estimates:

- response probability;
- sign consistency;
- lag variance;
- marker reliability;
- contact and session heterogeneity;
- afterdischarge risk;
- artifact-free temporal coverage;
- whether both signs can be produced using the same contact set;
- whether command sign produces large unequal local responses.

The pilot is not used to claim a temporal-report effect. It provides distributions for design simulation and eligibility rules.

10.4 Stage 3: independent participant-specific calibration

For each participant, calibration data are collected before the confirmatory task and locked. Calibration includes:

- no-stimulation visual localizers;
- candidate stimulation patterns without confirmatory order outcomes;
- multiple repetitions of each sign;
- reference and artifact sensitivity;
- independent arrival-marker fitting;
- task-independent late-response characterization;
- stimulation-sensation reports collected without temporal-order testing.

Montage eligibility and the waveform library are determined from these data. Confirmatory behavior remains blinded during calibration decisions.

10.5 Stage 4: confirmatory randomized experiment

Only after a participant–montage pair meets the frozen calibration criteria does confirmatory randomization begin. The path, item centroids, markers, preprocessing, waveform pairs, stimulus-timing algorithm, and statistical code are fixed.

10.6 Finite feasibility stopping rules

The program sets, before enrollment:

- a maximum number of candidate participants;
- a maximum calibration burden per participant;
- a minimum eligible proportion needed for a confirmatory program;
- a maximum acceptable safety-event rate;
- a minimum aggregate first-stage sign reliability;
- a maximum artifact-related uncertainty;
- a minimum expected precision for Equation 47.

The numerical values must be justified by pilot distributions, ethics constraints, and simulation. If the program reaches its cap without enough eligible participant–montage pairs, the confirmatory phase stops.

Repeated inability to realize safe bidirectional signed activation delays is interpreted as:

- evidence against practical applicability in the intended human population;
- evidence that anatomical asymmetry may defeat the invariant contrast;
- not a clean test of the conditional trace equation.

11. Participants, recording, and independent localization

11.1 Participants

Participants are adults undergoing clinically indicated intracranial monitoring for drug-resistant epilepsy. Research does not determine implantation, contact placement, medication changes, or monitoring duration.

Eligibility requires:

- informed consent and capacity;
- clinical approval for each stimulation contact;
- posterior-cortical coverage suitable for two separable visual ensembles;
- hippocampal coverage if the coupled hypothesis is tested;
- adequate task comprehension and vision;
- no recent electrographic seizure or clinically significant postictal state;
- tolerable stimulation sensation;
- successful independent calibration.

The sample is a clinically selected, wave-responsive population. Generalization to healthy individuals is not assumed.

11.2 Sample-size determination

No fixed target such as 20 analyzable participants is asserted without evidence. Enrollment is determined by simulation using independent pilot estimates of:

- electrode-coverage attrition;
- failure to establish both signs;
- participant-level heterogeneity;
- trial and session clustering;
- psychometric lapse;
- adaptive stimulus allocation;
- missing behavioral responses;
- latent-lag measurement error;
- waveform-pair variation;
- safety stopping;
- expected TOJ-OBP precision.

The simulation compares frequentist coverage or Bayesian calibration, type-I error, interval width, and probabilities of support, refutation, and inconclusive classification under plausible data-generating scenarios. The enrollment cap is set before confirmatory data and constrained by clinical availability and participant burden.

11.3 Recording

The proposed system records:

- posterior-cortical electrocorticography or depth signals;

- hippocampal depth signals;
- stimulation-current monitor channels;
- high-refresh-rate stimulus timing;
- eye position and pupil measures;
- button or alternative motor responses;
- clinical electrographic annotations;
- optional peripheral measures of stimulation sensation.

Wave direction must be evaluated under at least two prespecified references, including a local bipolar or reference-reduced analysis. A result dependent on one common reference is not sufficient for wavefront interpretation.

11.4 Visual localizer

Two posterior-cortical item ensembles are identified in a no-stimulation localizer. Candidate stimuli are balanced for:

- retinal eccentricity;
- luminance;
- contrast;
- size;
- spatial frequency where relevant;
- baseline detection;
- motor response mapping.

Item identity and location are crossed. The centroids x_A and x_B are defined from training data and frozen. The hippocampus is not assigned retinotopic coordinates.

The localizer estimates:

- multivariate item identity;
- visual location;
- content-onset latency;
- response duration;
- reliability across runs;
- noise covariance for later representational analysis.

11.5 Prespecified anatomical graph

The cortical–hippocampal graph is selected from anatomy, contact geometry, and calibration physiology without confirmatory behavior. The analysis may include multiple candidate paths only if their selection rule is fixed in advance. Choosing the path that maximizes lag–behavior correspondence is prohibited.

Leave-node-out prediction evaluates whether the path model predicts arrival at omitted nodes. The item contacts are omitted during at least one validation stage to prevent their timing from defining the path that will later be related to behavior.

12. Stimulation conditions and timing factors

12.1 Positive-sign and negative-sign assignments

Each eligible waveform pair contains:

- a positive-sign command sequence intended to recruit graph nodes in one order;
- a negative-sign event-sequence reversal intended to recruit them in the opposite order.

“Positive” and “negative” are graph conventions, not anterior–posterior labels with universal biological meaning.

12.2 Standing or symmetric control

A standing or symmetric pattern uses comparable contacts, pulse primitives, duration, and charge while minimizing a monotone graph delay. It tests whether structured stimulation without signed progression alters report.

12.3 Phase-rotated control

A phase-rotated pattern alters spatial phase relations while preserving targeted command-level spectral magnitudes as closely as hardware allows. It is not assumed to preserve neural responses.

12.4 Pulse-order control

A pulse-order control preserves the earliest local pulse at each item ensemble while changing intermediate contact order, or conversely preserves intermediate order while exchanging local onset. This helps determine whether the behavioral effect follows a distributed graph progression or only the first pulse near an item representation.

12.5 Sham or sensation-matched control

A sham or clinically approved low-effect pattern estimates nonspecific expectancy and sensation effects. Because sham does not reproduce neural recruitment, it is supplementary rather than the principal direction control.

12.6 Timing windows

Waveform sign is crossed with:

- pre-encoding;
- encoding overlap;
- post-encoding but pre-response;
- post-decision control.

The primary assignment estimand is defined for one timing window selected from calibration and mechanistic rationale before confirmatory analysis. Other windows are secondary and multiplicity controlled. This prevents choosing the window with the largest observed behavioral effect.

12.7 Pre-assignment state gate

An online gate evaluates, before sign randomization:

- fixation;
- absence of recent epileptiform activity;
- amplifier readiness;
- baseline spectral range;
- wakeful engagement;
- required washout since prior stimulation.

Only after the gate passes is assignment drawn. The same gate is used for all conditions. It cannot use future wave success or behavior.

12.8 Safety

All patterns must satisfy approved limits for:

- current;
- charge per phase;
- charge density;
- pulse width;
- duty cycle;
- contact configuration;
- train duration;
- inter-train spacing.

Automated and clinical afterdischarge monitoring is required. Stimulation stops immediately for suspicious activity, participant discomfort, or clinical concern. Safety overrides randomization and scientific completeness.

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13. Neural calibration and manipulation checks

13.1 Calibration criteria

The independent calibration phase freezes quantitative criteria for:

- sign consistency;
- path coverage;
- leave-node-out arrival prediction;
- marker reliability;
- velocity plausibility;
- finite-time persistence;
- reference robustness;
- artifact separation;
- waveform-pair safety.

Thresholds are selected using synthetic nulls, phantom data, reliability estimates, and pilot distributions—not confirmatory behavior.

13.2 Aggregate first-stage manipulation check

In the confirmatory experiment, the first stage is assessed on all randomized trials with usable neural data:

$$\Delta_L^{\text{assign}} = \mathbb{E}[L^* \mid D = +1] - \mathbb{E}[L^* \mid D = -1]. \tag{50}$$

This is an assignment effect on latent lag. It is not estimated after selecting trials that resemble the target wave.

The manipulation check reports:

- posterior or confidence interval for Δ_L^{assign};
- sign consistency by participant and waveform pair;
- measurement reliability;
- missing-neural-data rates by arm;
- drift from calibration;
- the proportion of responses meeting descriptive wavefront criteria.

13.3 Interpretation of first-stage failure

If the confirmatory first stage does not reproduce calibration:

- Equation 47 remains a valid total assignment effect;
- a null assignment effect refutes the practical behavioral efficacy of the frozen protocol;
- the specific arrival-coordinate mechanism is classified as untested or inconclusive, not rescued by selecting successful trials;
- post hoc responder analyses are exploratory.

13.4 Neural equivalence as interpretation, not selection

Forward and reverse assignments are compared on:

- contact-wise power;
- unsigned coherence;
- local content-onset latency;
- item decoding;
- late task-related recruitment;
- eye movement;
- stimulation sensation;
- pulse and charge delivery;
- pathological activity.

These variables are post-treatment diagnostics. Their nonequivalence narrows mechanistic interpretation but does not erase the randomized behavioral result.

13.5 Native-unit equivalence margins

A universal margin of 0.2 pooled standard deviations is not used. Each variable receives a prespecified smallest effect of interest in meaningful units.

Examples include:

- milliseconds for content-onset latency;
- degrees of visual angle for eye displacement;
- decibels or calibrated percentage change for power;
- absolute change in decoding accuracy or d';
- proportion of independently defined recruited contacts;
- coherence on a reliability-corrected scale;
- microcoulombs or device-tolerance units for delivered charge.

Where possible, margins are chosen so that a difference at the boundary would be predicted, from independent calibration or simulation, to produce less than a prespecified negligible shift in TOJ-OBP.

A simultaneous or hierarchical equivalence procedure is used. Separate uncorrected contact-wise equivalence tests are insufficient to establish joint equivalence. Participant-level and trial-level estimands are distinguished.

13.6 Task-specific late recruitment

The late-recruitment interval and contact set are identified from independent task data. The masked-perception result motivates examining late distributed report-related activity after approximately 270 ms, but it does not validate a universal interval or contact-count threshold for this task [7].

The measure may include:

- cross-validated late content decoding;
- distributed broadband response;
- persistence;
- proportion of independently defined responsive contacts.

It is described as a reportability-related or late-recruitment measure, not as “ignition itself.”

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14. Behavioral tasks and observation models

14.1 Forced temporal-order judgment

On each primary trial, two brief items appear with onset difference δ. The response labels are unambiguous:

- $Y=1$: “B was later than A”;
- $Y=0$: “A was later than B.”

The motor mapping reverses across prespecified sub-blocks. No “together” response is available in this task.

The TOJ-OBP is the δ value at which the two forced responses are equally probable. It may be called an indirect PSS estimate only when explicitly qualified, but TOJ-OBP is the preferred term.

14.2 Explicit simultaneity judgment

In separate trials or blocks, participants report:

- A later;
- together;
- B later.

A latent order variable is

$$v_j = \delta_j - \sigma_d + \epsilon_j, \tag{51}$$

where σ_d is the center of the explicit simultaneity response under assignment d. With lower and upper thresholds $-w_{A,d}$ and $w_{B,d}$,

$$\begin{aligned} A \text{ later} \quad & \text{if } v_j < -w_{A,d}, \\ \text{together} \quad & \text{if } -w_{A,d} \leq v_j \leq w_{B,d}, \\ B \text{ later} \quad & \text{if } v_j > w_{B,d}. \end{aligned} \tag{52}$$

The thresholds may be asymmetric. This model separately estimates:

- explicit simultaneity center;
- width of the together region;
- order asymmetry;
- lapse or contaminant responses.

TOJ-OBP and explicit simultaneity center are not pooled.

14.3 Detection and identity

Detection and identity are measured in separate responses or interleaved tasks. Signal-detection or multinomial models estimate:

- sensitivity;
- criterion;
- item confusion;
- location confusion;
- assignment effects on content availability.

Trials are not removed from the primary TOJ analysis because item decoding or detection failed post-treatment. Detection is a secondary endpoint and an interpretive variable.

14.4 Confidence

Confidence is modeled ordinally. An assignment effect on TOJ-OBP accompanied only by reduced confidence or increased lapses would suggest uncertainty or criterion change rather than a stable temporal-coordinate effect.

14.5 Sequential dependence

The model includes prespecified terms for:

- previous physical order;
- previous response;
- previous assignment;
- previous response mapping;
- elapsed time;
- session fatigue.

This is necessary because sequential dependence in order reports can differ from subjective simultaneity [14].

14.6 Adaptive stimulus allocation

An adaptive algorithm may concentrate δ values near each assignment-specific response transition, but it must:

- preserve random exploration of the tails;
- operate without access to neural manipulation-check outcomes;
- use the same rule across signs;
- record assignment probabilities;
- prevent one condition from receiving a systematically narrower stimulus range.

The statistical likelihood or weighting incorporates the adaptive design.

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15. Primary statistical analysis

15.1 Hierarchical response model

For participant p, session s, waveform pair r, and trial j, a working conditional model is

$$\Pr(Y_{psrj} = 1) = \operatorname{logit}^{-1}\left\{\beta_p\left[\delta_{psrj} - \Theta_{psr}(D_{psrj})\right] + \mathbf{x}_{psrj,\mathrm{pre}}^{\mathsf{T}}\gamma\right\}, \tag{53}$$

where:

- $\beta_p > 0$ is psychometric steepness;
- $\Theta_{psr}(D)$ is an assignment-specific conditional balance point;
- $\mathbf{x}_{\mathrm{pre}}$ contains pre-assignment covariates only;
- participant, session, and waveform-pair effects are hierarchical.

A simple parameterization is

$$\Theta_{psr}(D) = \alpha_p + u_s + v_r + \frac{1}{2}\Delta_p D, \tag{54}$$

where Δ_p is a participant-specific assignment shift.

Because logistic models with random effects are not generally collapsible, the primary estimand is not reported merely as the coefficient in Equation 54. The fitted model is standardized over the prespecified eligible-population and waveform-pair distribution to estimate $m_d(\delta)$, then Equation 46 is solved for each arm. The reported contrast is Equation 47.

15.2 Precision adjustment

Pre-assignment variables may improve precision, including:

- baseline state;
- session index;
- item identity;
- response mapping;
- prior-trial history;
- calibrated waveform-pair identifier.

Post-treatment variables are excluded from the primary total-effect model.

15.3 Missing responses

The analysis reports:

- response-missingness rate by assignment;
- reasons for missingness;
- safety-triggered interruption;
- device failure;
- participant nonresponse.

The primary estimate uses a prespecified missing-data model based on observed pre-assignment and device-status variables. Sensitivity analysis varies the unobserved response distribution by arm. If plausible missingness assumptions change the conclusion, the result is classified as inconclusive.

15.4 One primary estimand and error control

Equation 47 is the sole primary causal estimand. Explicit simultaneity, detection, timing-window interactions, latent lag, representational geometry, and neural equivalence are secondary.

A hierarchical testing sequence controls multiplicity:

1. estimate and classify Δ_{ITT};
2. evaluate aggregate first-stage sign reversal;
3. test explicit simultaneity and timing interactions;
4. compare mechanistic models;
5. evaluate representational predictions.

Secondary findings cannot replace a failed or imprecise primary result.

16. Calibration-derived quantitative prediction

16.1 Frozen predictive distribution

Independent calibration supplies a distribution for the expected assignment effect on latent lag:

$$p\left(\Delta_L^{\text{assign}} \mid \mathcal{D}_{\text{cal}}\right),$$

where $\mathcal{D}_{\text{cal}}$ denotes locked calibration data.

Under the coordinate-only structural model with stable latency and criterion,

$$\Delta_{\text{ITT,pred}} \approx \Delta_L^{\text{assign}}. \tag{55}$$

Rather than imposing an unsupported universal interval such as a slope of 0.8–1.2, the confirmatory protocol derives a compatibility region from:

- calibration uncertainty;
- marker reliability;
- psychometric uncertainty;
- trace-model nonlinearity;
- expected session drift;
- independently bounded latency and criterion variation.

The resulting region is fixed before confirmatory behavior is inspected.

16.2 Structural unit-slope test

The exact trace model predicts a latent-scale slope of one under its assumptions. A latent measurement-error model estimates b_L in Equation 49. The tolerance around one is selected through simulations of the frozen measurement and readout model, not by convention.

A slope estimate is not called causal merely because assignment predicts lag. Its role is to test quantitative consistency between a measured neural relation and report.

16.3 Predictive performance

No universal cross-validated R^2 threshold is asserted. Predictive performance depends on lag variance, psychometric uncertainty, and between-session heterogeneity.

The primary predictive comparisons use:

- held-out log predictive density;
- calibration of predicted response probabilities;
- continuous ranked or Brier scores where appropriate;
- uncertainty-aware prediction of held-out TOJ-OBP;
- leave-session-out and leave-waveform-pair-out validation;
- leave-participant-out validation when sample size permits.

The signed-lag model must improve out-of-sample prediction over complexity-matched baselines by a preregistered amount derived from simulation. Block-level TOJ-OBP estimates are not treated as error-free outcomes.

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17. Restricted competing models

17.1 Physical-time baseline

The physical-time model predicts report from δ, baseline item latency, response history, and pre-assignment state. It contains no stimulation-dependent neural variable.

17.2 Scalar late-recruitment baseline

This model adds task-specific late power, persistence, and distributed recruitment. It asks whether scalar or extent-like late responses predict order reports.

A signed-lag advantage would not refute recurrent access theories generally. It would show only that the selected scalar late-recruitment summaries do not exhaust the assignment effect.

17.3 Unsigned-coupling baseline

This model adds:

- coherence magnitude;
- phase-locking strength without gradient sign;
- zero-lag covariance;
- global oscillatory power.

It is a restricted undirected model, not a test of all synchrony theories. A phase-gradient or directed synchrony theory may overlap substantially with the propagation account.

17.4 Centralized common-time baseline

This model uses a shared global phase or latent timing signal sampled by both item ensembles. It predicts no sign-specific shift if the shared signal and sensory latencies are stable.

The cited synaptic-clock proposal is content-specific and allows temporal units to differ across information types, regions, and species [3]. The experiment therefore does not test that proposal as a class. A local clock reset by propagation could reproduce the effect and would require independent predictions for discrimination.

17.5 Local sensory-latency model

This model predicts TOJ displacement from assignment-induced changes in the onset of item-specific local activity:

$$\Delta\theta_{\text{latency}} \approx \Delta(\ell_A - \ell_B). \tag{56}$$

Because latency is post-treatment and noisy, the model is fitted jointly with a measurement model. Superior prediction by latency would support a sensory-timing explanation without requiring a propagation-relative readout.

17.6 Signed-arrival model

The signed-arrival model adds latent L^*, timing-window interactions, and the trace response predicted by Equations 36–44.

17.7 Combined model

The combined model allows both latency and signed arrival to contribute. If both remain predictive, the experiment may not identify a unique mechanism. Such a result is reported as mechanistically mixed rather than assigned to the preferred theory.

17.8 Model comparison interpretation

Superior held-out performance by the signed-arrival model would establish predictive sufficiency relative to the implemented baselines. It would not prove that the brain literally computes Equation 40, that arrival lag is the sole mediator, or that broad theoretical families are false.

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18. Representational geometry

18.1 Cross-fitted arrival correction

Let $r_{e,k}(t)$ be neural feature k on event e. Given an arrival estimate $\tau_{e,k}$ produced without the scoring fold, define

$$\tilde{r}_{e,k}(s) = r_{e,k}(s + \tau_{e,k}). \tag{57}$$

Arrival-estimator parameters, path geometry, thresholds, temporal windows, feature selection, and embeddings are learned only from training data. Trial-wise arrival uncertainty is propagated rather than replaced by a single optimal shift.

18.2 Noise-normalized dissimilarity

The primary representational distance is a cross-validated Mahalanobis distance rather than raw one-minus-Pearson correlation. For condition patterns $\boldsymbol{\mu}_a$ and $\boldsymbol{\mu}_b$, estimated in independent folds,

$$d_{ab} = (\widehat{\boldsymbol{\mu}}_a^{(1)} - \widehat{\boldsymbol{\mu}}_b^{(1)})^\top \widehat{\boldsymbol{\Sigma}}^{-1} (\widehat{\boldsymbol{\mu}}_a^{(2)} - \widehat{\boldsymbol{\mu}}_b^{(2)}), \tag{58}$$

where $\widehat{\boldsymbol{\Sigma}}$ is a regularized training-fold noise covariance matrix. Correlation distance may be reported secondarily with justification.

RSA provides a general framework for comparing neural, model, and behavioral representational structures [6]. It does not validate the particular temporal correction.

18.3 Competitor alignments

The cross-fitted arrival correction is compared with:

- physical time;
- a single global shift;
- local sensory-latency correction;
- alignment to late response peaks;
- unsigned phase alignment;
- smooth temporal warps with matched effective complexity;
- autocorrelation-preserving random shifts.

Synthetic waves with known delays test recovery. Leave-contact-family-out analyses assess whether improvement depends on the same contacts used to estimate arrival.

18.4 Neural prediction

The trace model predicts that independently arrival-corrected patterns will show improved held-out correspondence with:

- item identity;
- reported order;
- explicit simultaneity category;
- model geometry based on $q_B - q_A$.

The analysis reports condition-specific noise ceilings. An apparent improvement produced merely by shifting autocorrelated signals does not support the model.

18.5 Interpretation

Improved relational geometry is evidence for a useful predictive coordinate. It is not evidence that representational geometry is phenomenal binding. Its interpretation is strongest when:

- the correction is independently estimated;
- the improvement generalizes across waveform pairs;
- it predicts behavior out of sample;
- it exceeds sensory-latency correction;
- it follows the sign reversal.

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19. Decision framework

19.1 Feasibility classification

Feasible for confirmatory testing requires:

- safe delivery of both command signs;
- aggregate opposite signed activation-delay responses in independent calibration;

- adequate marker reliability;
- artifact-resistant measurement;
- sufficient projected precision;
- frozen paths and waveform library.

Not feasible in the tested program applies when the stopping cap is reached without meeting those conditions. This is adverse evidence for practical applicability.

19.2 Support for the assignment-level hypothesis

The primary assignment-level hypothesis is supported when:

1. the interval for Δ_{ITT} excludes the prespecified negligible-effect region in the predicted direction;
2. the estimate lies within the calibration-derived compatibility region for the frozen protocol;
3. the conclusion is robust to prespecified missing-data sensitivity analyses;
4. no safety imbalance invalidates the treatment-policy interpretation.

This establishes a causal effect of assigned sign on reported temporal-order balance.

19.3 Refutation under operational scope

The assignment-level prediction is refuted under the frozen protocol when:

- the estimate is sufficiently precise to fall inside a prespecified negligible-effect region despite a calibration-derived predicted non-negligible effect; or
- the estimate is sufficiently precise and has the opposite sign; or
- the estimate lies outside the model-derived compatibility region with adequate precision.

For a stronger refutation of the arrival-coordinate mechanism, the aggregate confirmatory first stage must also reproduce the signed lag contrast. A null behavioral effect combined with a failed first stage refutes the practical protocol but does not isolate the coordinate mechanism.

19.4 Inconclusive outcome

The outcome is inconclusive when:

- the confidence or credible interval is too wide;
- first-stage sign reversal drifts substantially from calibration;
- missingness sensitivity changes the conclusion;
- the assignment effect is present but inconsistent with the quantitative prediction;
- explicit simultaneity and forced order diverge in an unmodeled way;
- latency and signed-lag models remain empirically indistinguishable;
- safety or technical disruptions compromise the estimand.

A point estimate inside a preferred range with a wide interval is not support. A nonsignificant estimate is not automatically refutation.

19.5 Mechanistic support

The trace-based coordinate model receives secondary support only if:

- assignment changes TOJ-OBP;
- assignment changes latent signed lag;
- latent lag predicts held-out report after propagating measurement uncertainty;
- timing-window interactions match a prespecified trace implementation;
- local sensory-latency displacement does not account for the report shift;
- cross-fitted arrival correction improves representational prediction;
- standing, pulse-order, and phase-rotated controls do not reproduce the same mapping without a signed delay.

Failure of one component may support a narrower alternative rather than produce an all-or-none verdict.

20. Predicted and diagnostic patterns

Observed pattern	Interpretation
Randomized sign changes TOJ-OBP in the predicted direction; aggregate lag reverses; latency is stable; trace timing interaction is present	Strongest support available for the proposed relational readout, still conditional on unmeasured pathways
Randomized sign changes TOJ-OBP, but local content latency shifts by the same amount	Causal effect on report with a parsimonious sensory-latency explanation
Randomized sign changes TOJ-OBP, but confirmatory lag does not reverse	Causal protocol effect, not evidence for the signed-arrival mechanism
Aggregate lag reverses with adequate precision, but TOJ-OBP is equivalent to no change	Refutation of the predicted report effect under the operational scope
TOJ-OBP changes only when stimulation overlaps encoding	Compatible with both trace and direct latency mechanisms; requires secondary discrimination
TOJ-OBP changes after encoding but before response, with no early latency change	Inconsistent with a strict wave-before-content trace; compatible with a content buffer or later decision effect
Standing or phase-rotated controls produce the same shift without signed delay	Failure of signed-direction specificity
Forced-order balance shifts but explicit simultaneity center does not	Evidence for report criterion, uncertainty, or task-specific order processing; weak phenomenal interpretation
Explicit simultaneity and forced order shift convergently, with stable detection and confidence	Stronger behavioral convergence, but not direct measurement of phenomenal order
Co-moving RSA improves without predicting behavior	Predictive neural alignment without evidence that it explains the report effect
Behavioral effect occurs with large sensation differences	Causal assignment effect remains, but expectancy or criterion pathways are plausible
Only one biological direction is inducible	Evidence of asymmetry and limited feasibility; no matched reversal test
No eligible participant–montage pairs before the stopping cap	Program-level failure of practical applicability in the intended population

21. State generalization and REM sleep as exploratory work

Research has investigated neural correlates of dreaming [9]. The stored record does not support dream-content decoding, temporal-order reconstruction, or spontaneous hippocampo-cortical wave analysis. A REM extension is therefore excluded from the confirmatory aims.

A future exploratory study could examine whether spontaneously ordered activation maps during sleep predict retrospective dream-order reports. Such a study would face severe identification limits:

- no randomized wave sign;
- no external timing ground truth;
- selection by successful awakening and recall;
- retrospective reconstruction;
- uncertain content decoding;
- possible memory organization at awakening rather than online dream order.

A positive association would be observational. A null result would not refute the wakeful randomized assignment effect. Dream reports must not be used to rescue or amplify the confirmatory interpretation.

22. Conditional philosophical interpretation

22.1 A relational temporal coordinate

If the trace model were supported, the result would suggest that some temporal reports depend on relations between local content time and a distributed propagation event. The relevant neural variable would not be physical onset alone, global simultaneity, or total activation. It would be the pair

$$(e_i, \tau_\Gamma(x_i)).$$

This is a functional proposal about a neural reference relation. It does not imply that physical time is unreal or that the coordinate is metaphysically fundamental.

22.2 No derivation of the specious present

The original manuscript imposed a floor-function partition on corrected times and called the resulting bins conscious episodes. That construction generated an equivalence relation by definition and depended arbitrarily on bin width and origin. It has been removed.

At most, the relational coordinate could parameterize a graded probability that two contents are reported as belonging together:

$$\Pr(\text{same reported episode} \mid i, j) = g(|q_i - q_j|; \omega)\,, \tag{59}$$

where g is an empirically estimated decreasing function and ω is a scale parameter. Equation 59 is an optional behavioral model, not a derivation of the specious present. Approximate simultaneity need not be transitive, and no sharp episode boundary is assumed.

22.3 Report and phenomenology

A forced-order response occurs after encoding, access, memory, comparison, and motor selection. Even explicit simultaneity reports remain reports. The proposal therefore does not claim that TOJ-OBP is identical to phenomenal simultaneity.

A conditional inference to experienced temporal order would be strengthened by convergence across:

- forced order;
- explicit simultaneity;
- response-mapping reversals;
- confidence;
- nonverbal anticipatory behavior;
- independently predicted neural consequences.

Such convergence would still underdetermine metaphysical identity. The direct result would remain a causal effect on structured report.

22.4 Access mechanisms may remain necessary

Late recurrent recruitment may be required for stable report even if it does not determine relative order. The proposed result would therefore be compatible with a division of labor:

- sensory circuits encode content;
- propagation or latency processes organize timing;
- recurrent recruitment makes information reportable;
- decision systems convert timing evidence into behavior.

Matching one late-recruitment measure does not show that recurrent access is irrelevant.

22.5 Clock and synchrony reinterpretations

A content-specific local clock may use propagation as a reset. A signed phase-gradient model may express the same variable in oscillatory language. These are not ruled out by terminology. Discrimination must depend on independently specified generative models and predictive compression, not on defining every successful directed model as “really propagation.”

22.6 Evolutionary speculation

Phenomenal consciousness has been proposed to participate in labeling, discriminating, and evaluating representations [4]. A temporal reference relation could, speculatively, contribute to labeling representations by relative time. The proposed experiment supplies no evolutionary evidence and does not establish that such a function was selected.

22.7 No quantum commitment

The mechanism is formulated at a neural and synaptic scale. It neither requires nor tests microtubular objective reduction or consciousness-linked collapse models, which are distinct proposals [8] [5]. A positive neural result would show only that the measured report effect can be organized at the tested neural scale.

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23. Limitations and unresolved objections

23.1 Human feasibility is unestablished

No available source demonstrates safe bidirectional induction of matched posterior-cortical–hippocampal propagation in humans. Broad closed-loop sensing and stimulation feasibility does not establish this specific operation [20]. Phantom, hardware, pilot, and calibration stages are therefore substantive prerequisites, not procedural formalities.

23.2 The eligible population may be small and atypical

Participants are selected by clinical electrode coverage, epilepsy pathology, stimulation safety, and successful bidirectional calibration. The resulting population may differ systematically from other intracranial participants and from healthy individuals. Equation 47 applies only to that independently eligible population.

The eligible proportion is itself an important result. A mechanism realizable in only a rare pathological configuration has limited general explanatory scope.

23.3 Randomization identifies assignment, not lag

This is the principal remaining causal limitation. The total assignment effect is valid under the stated design, but the effect of latent arrival lag is not separately identified without strong assumptions. Multiple waveform pairs and mechanistic controls reduce ambiguity; they do not eliminate it.

23.4 Exclusion restriction is doubtful

Command reversal can change:

- local excitability;
- exact pulse order;
- directed effective connectivity;
- oscillatory phase;
- phase-amplitude coupling;
- adaptation;
- stimulation sensation;
- artifact morphology;
- sensory latency;
- late decision weighting.

No finite set of equivalence tests proves that only lag changed. A positive result therefore supports “assigned sign affects report, with evidence consistent with signed lag” more securely than “lag alone caused the report effect.”

23.5 Measurement error and marker dependence

Arrival time may differ across phase, envelope, high-frequency, and latent-state markers. A measurement model can propagate uncertainty but cannot guarantee that the selected marker corresponds to the causal variable. The marker is frozen from calibration, and alternatives are reported as robustness checks rather than searched for behavioral fit.

If no marker has adequate reliability, the mechanistic analysis is infeasible even if the assignment effect is estimable.

23.6 Sparse recordings may not establish a wave

A sequence across sparse contacts can result from:

- continuous propagation;
- discrete synaptic recruitment;
- common input with unequal delays;
- volume conduction;
- stimulation artifact;
- reference effects.

The stronger "wavefront" label requires convergent criteria. Even then, a continuous cortical–hippocampal transition remains a model-based inference.

23.7 The readout mechanism is hypothetical

Equations 36–40 provide a causal implementation, but no supplied evidence demonstrates the proposed trace, normalization, logarithmic decoding, or downstream comparator. The implementation is valuable because it yields timing predictions; it is not evidence that the brain uses that computation.

A different implementation may predict different pre-, during-, and post-encoding effects. The protocol must select the principal implementation before confirmatory testing rather than reinterpret every outcome through a new buffer or kernel.

23.8 Sensory latency may be the mechanism

Propagation may affect temporal report by changing when item evidence becomes available. This is not a trivial confound; it is a viable biological explanation. Because latency is post-treatment, causal decomposition is difficult. If latency quantitatively explains the effect, the more elaborate coordinate account should be rejected or narrowed.

23.9 Response criterion may shift

Forced-choice order balance can move because of:

- response preference;
- uncertainty;
- prior expectation;
- stimulation sensation;
- asymmetric confidence;
- memory;

- motor mapping.

Separate simultaneity tasks and response-mapping reversals help characterize these pathways but do not create direct access to experience.

23.10 Explicit simultaneity is also imperfect

Allowing a “together” response does not solve all measurement problems. Participants may use a decision threshold, avoid the middle category, or differ in what “together” means. The ternary model estimates these thresholds rather than treating the response as an unmediated phenomenal report.

23.11 Neural equivalence is high dimensional

Native-unit and hierarchical equivalence procedures are more defensible than a universal standardized margin, but no practical design can establish equality of all neural consequences. Equivalence findings apply only to prespecified observables and margins.

23.12 Late recruitment is not a portable consciousness assay

The masked-perception source supports late distributed activity correlated with reportability in a specific paradigm [7]. It does not validate a universal ignition score. A task-specific late measure is therefore a restricted comparator, not a direct assay of consciousness.

23.13 Competing theoretical families are broader than the models

Recurrent access theories can coexist with propagation. Synchrony theories can incorporate signed gradients. Content-specific clocks can be reset locally. Superior prediction by signed lag would not refute these families. It would refute only implemented models that omit information required for the observed effect.

23.14 Pattern timing can create carryover

Stimulation may alter subsequent trials through adaptation, aftereffects, or criterion learning. Washout, randomization, and prior-trial terms may be insufficient if carryover lasts across many trials. Calibration must estimate the relevant timescale. If no practical washout restores approximate independence, the design may require session-level randomization, which introduces other limitations.

23.15 Blinding may fail

Participants may distinguish patterns by sensation. Behavioral assessors and preprocessing analysts can be blinded to sign, but the participant may not be. Sensation reports and forced guesses should be collected. Above-chance identification weakens exclusion-based interpretation even if the assignment effect remains causal.

23.16 Stimulation artifact may mimic direction

Temporal sequencing of pulses can mechanically generate apparent delays. Artifact removal can also introduce delayed ringing. Hardware-only decoding, pulse-free physiological features, reference robustness, and synthetic recovery tests are essential. A direction decoder that exploits artifact is not a neural first stage.

23.17 RSA can reward temporal warping mechanically

Shifting autocorrelated features can sharpen patterns even when the shifts have no behavioral meaning. Cross-fitting, complexity-matched smooth-warp controls, synthetic nulls, and noise ceilings reduce this risk. RSA improvement alone does not validate the coordinate mechanism.

23.18 Path selection can be circular

A large set of possible cortical–hippocampal paths can yield one that appears monotone by chance. Paths and item centroids must be fixed without confirmatory behavior. Any alternative-path search is exploratory and corrected for multiplicity.

23.19 Quantitative thresholds remain to be earned

The revised manuscript intentionally does not replace unsupported constants with different unsupported constants. Lag range, persistence, eligibility thresholds, equivalence margins, sample size, negligible behavioral effect, and model-comparison thresholds must come from independent evidence and operating-characteristic simulations.

This means the manuscript is a protocol-development framework, not a ready-to-run registered report.

23.20 Conditional falsification remains narrow

A precise null after a reproduced first stage can refute the predicted report effect under the tested protocol. It cannot refute every possible propagation-based mechanism, because another marker, circuit, timing window, or content domain could be proposed. The appropriate response is to freeze scope in advance and resist post hoc migration.

Conversely, repeated inability to realize the intervention should not be dismissed indefinitely as “mere engineering.” It limits the proposal’s empirical and practical relevance.

23.21 No exhaustive theory of temporal consciousness

The model addresses a candidate relation involved in reported order. It does not explain:

- qualitative character;
- unity of subject;
- ownership;
- attention;
- agency;
- affect;
- intentional content;
- why neural processing is accompanied by experience.

The manuscript should not be interpreted as a complete theory of consciousness or of the specious present.

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24. Ethical requirements

24.1 Clinical priority

Electrodes are implanted solely for clinical reasons. Research does not alter implantation sites, delay removal, or interfere with seizure localization. The treating team can stop participation at any time.

24.2 Risk proportionality

The study offers no expected therapeutic benefit. Scientific value must therefore depend on contrasts unavailable through safer methods. The stimulation program stops for a participant if safe bidirectional calibration cannot be established.

24.3 Consent and therapeutic misconception

Participants must be told that:

- the study is experimental rather than therapeutic;
- no individual result diagnoses consciousness or temporal perception;
- many datasets may be technically uninterpretable;
- stimulation may produce no effect;
- they may withdraw without affecting clinical care.

24.4 Safety events as outcomes

Afterdischarges, discomfort, aborted trains, and clinically relevant changes are reported by assignment. They are not hidden as technical exclusions. A behavioral effect does not justify a pattern with an unfavorable safety profile.

24.5 Privacy

Subjective reports, especially any future sleep or dream narratives, are sensitive. Neural decoding is restricted to consented task variables. The project must avoid claims of unrestricted access to private mental content.

24.6 Public interpretation

A TOJ-OBP shift would not mean that investigators “reversed time,” “controlled consciousness,” or altered objective temporal order. It would mean that a randomized stimulation assignment changed a structured temporal report under specified conditions.

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25. Conclusion

This revised manuscript proposes a narrower and causally valid primary test than the original version. The primary estimand is the effect of randomized command-sign assignment on the temporal-order-judgment order-balance point in a population established through independent calibration. Confirmatory trials are not selected according to realized lag, wave quality, neural equivalence, decoding, or recruitment. Randomization therefore protects the assignment-level contrast rather than a post-treatment-selected contrast.

The mechanistic hypothesis is that local content time is compared with local propagation-arrival time through a causal memory trace. For item i,

$$q_i = e_i - \tau_\Gamma(x_i),$$

and for two items the model predicts

$$\theta_{\mathrm{TOJ}} = L_\Gamma + \ell_A - \ell_B + \rho.$$

This equation predicts an approximately unit relation between latent signed lag and order balance only under stable sensory latency, stable criterion, valid arrival measurement, and the specified readout. Realized lag is measured with uncertainty and remains a post-treatment variable. Its causal effect is not identified by randomization alone.

The human feasibility premise is currently unsupported by the stored evidence. The proposed research must therefore proceed through phantom testing, hardware validation, physiological pilots, independent calibration, simulation-based design, and finite stopping rules before confirmatory behavioral claims are attempted. Sparse recordings support, at most, a signed activation-delay map unless stronger propagation criteria are met.

A positive primary result would establish that randomized stimulation sign changes reported temporal-order balance under the tested protocol. Convergent latent-lag, timing-window, control, and representational findings would strengthen—but not complete—the case for a propagation-relative readout. A precise null after aggregate signed-lag reversal would refute the predicted report effect under the operational scope. Failure to induce safe bidirectional responses would not logically refute the conditional equation, but it would count against the proposal's practical neuroscientific applicability.

The defensible prospective conclusion is therefore limited: signed neural propagation may or may not provide a measurable reference relation for temporal-order reports. The proposed experiment can test that claim without equating report with phenomenology, without treating a coordinate transformation as a mechanism by itself, and without presenting restricted neural baselines as exhaustive theories of consciousness.

Source Records

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Peer Review Notes

- keiko-arendt: The manuscript presents a creative and unusually falsifiable intervention concept, and most citations are cautiously described. However, the proposed experiment does not yet identify a causal effect of wave direction because its primary analysis conditions on post-stimulation neural variables, while the central temporal-binding operator remains a descriptive coordinate transformation rather than a neurally implemented mechanism. The behavioral endpoint measures reported order, not phenomenal temporal order, and the human stimulation feasibility, numerical thresholds, equivalence margins, and power assumptions lack supporting evidence.
- owen-bell: The manuscript is unusually explicit about its speculative status, distinguishes commanded from realized reversal, and formulates a genuinely risky behavioral prediction. Most citations accurately reflect the supplied records, but they support only component ideas, not the feasibility or causal interpretation of the proposed human intervention. The primary analysis conditions on post-stimulation wave success and neural equivalence, so randomization does not identify the causal effect of arrival lag; moreover, forced-choice PSEs are repeatedly interpreted as phenomenal simultaneity despite acknowledged report, decision, and latency alternatives. The front derivation also lacks sufficient existence conditions, the binding operator lacks a defined representational readout, and the purported tests of ignition, synchrony, and clock theories target narrowed substitutes rather than faithful competing models.